

SCOTT TRC-1 AIR CART — QUICK REFERENCE

Model 200070-01 | 6-Outlet Cart, Hansen Fittings | Source: Scott Safety O&M Instructions P/N 595269-01 Rev. A (6/12)
— this sheet is a summary; refer to the full manual for complete procedures and warnings.

WHAT IT DOES

Wheeled cascade cart supplying breathing air to respirators (SCBA extended-duration hoseline, Type C airline) from two onboard cylinders and/or a remote high-pressure line. Two separate manifolds: **Respirator Supply** (0–100 psi, breathing air only) and **Auxiliary Supply** (tools or backup). Supports up to 8 respirator users depending on configuration.

AIR SOURCE SELECTOR

Setting	Effect
Common Air Source	Both manifolds draw from whichever source is opened (cylinders or remote line).
Isolated Air Source	Respirator Supply = cylinders ONLY. Auxiliary Supply = remote line ONLY. Always use this setting when running pneumatic tools.

SETUP — CYLINDERS ONLY

- 1 Both outlet pressure regulators turned fully counterclockwise (zero) BEFORE opening any valve.
- 2 Confirm both cylinders full, cylinder valves closed, vent valves closed.
- 3 Slowly open one cylinder valve — alarm chirps briefly until pressure passes 1000 psi, then resets.
- 4 Turn Respirator Outlet Regulator and Auxiliary Regulator up to 60–100 psi with nothing connected; check for leaks/hiss.
- 5 Connect respirator hoses — confirm a firm "click" on each quick-disconnect; tug-test.

LOW-PRESSURE ALARMS

Whistle = cylinder supply low. **Bell** = remote high-pressure line low. Both trip at approx. **500 psi** and CANNOT be silenced — they sound continuously until pressure is restored above 1000 psi or the cylinder empties.

On alarm: open the reserve cylinder → close & vent the depleted one → swap in a fresh, fully-charged cylinder immediately.

HOT-SWAPPING A CYLINDER MID-USE

Users stay on air the whole time — always open the fresh cylinder BEFORE closing the depleted one.

- 1 When the whistle sounds (or gauge reads low), open the valve on the UNUSED cylinder.
- 2 Wait for inlet pressure to climb back above 1000 psi — the alarm resets itself automatically.
- 3 Now close the valve on the depleted cylinder (push in, turn clockwise).
- 4 Crack open that cylinder's vent valve to bleed the trapped line pressure.
- 5 Once vented, disconnect the high-pressure coupling (hand-tight only — no wrench), loosen the retention strap, and pull the empty cylinder.
- 6 Strap in a fresh, fully-charged cylinder, valve outlet oriented toward the coupling; confirm the gasket is present and undamaged; hand-tighten the coupling.

CONNECTING AN EXTERNAL SOURCE (AIR TRUCK / CASCADE)

The cart's High Pressure Inlet accepts a remote high-pressure hose line up to 5500 psi.

- The supply hose must have a shut-off valve at the outlet end.
- The external source must hold the respirator's minimum required outlet pressure all the way down to approx. 500 psi — verify this in a clean air area before relying on it.
- A check valve isolates the remote inlet from the onboard cylinders — truck air CANNOT be used to recharge the cart's two cylinders, it only feeds the manifolds directly while connected.
- Common Air Source: truck air feeds both Respirator and Auxiliary manifolds; cylinders sit as backup.
- Isolated Air Source: truck air is dedicated to Auxiliary only (e.g., running tools) while cylinders exclusively supply the Respirator manifold — use this setting any time the truck may also run pneumatic tools, so tool air never touches the breathing air path.

- Standard practice: keep two fully charged cylinders loaded as backup any time the cart is running off an external/truck source.

SHUTDOWN SEQUENCE

- 1 Confirm all respirator users are in a safe (non-hazard) area.
- 2 Users doff respirators and disconnect hoses from the cart.
- 3 Fully close both cylinder valves (push in, turn clockwise).
- 4 Close and disconnect remote high-pressure line, if used.
- 5 Slowly open both vent valves to bleed remaining system pressure.
- 6 Vent trapped air: partially engage a male quick-disconnect into one Respirator outlet and one Auxiliary outlet.
- 7 Turn both outlet pressure regulators fully counterclockwise (zero).
- 8 Move cart to safe area; clean, inspect, and re-rack per Maintenance section.

CRITICAL SAFETY RULES

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| ■ NEVER connect tools to Respirator Outlets — lubricant contamination risk. Tools go on Auxiliary Outlets only. |
| ■ Once Auxiliary Outlets have supplied tools, they are NOT to be used for respirators again until air quality is verified clean (Grade D, CGA G-7.1 or better). |
| ■ NOT rated for IDLH atmospheres unless the respirator itself carries an emergency escape/backup air supply. |
| ■ A trained attendant with two-way comms MUST monitor the cart at all times while it's supplying respirators. |
| ■ Hand-tighten high-pressure couplings ONLY — never use a wrench (damages the coupling gasket). |
| ■ Air tools can rapidly deplete cylinder air — always use Isolated Air Source when tools are running so breathing air is protected. |
| ■ Use only clean breathing air (CGA G-7.1 Grade D+, dew point ≤ -65°F) — never oxygen or shop air. |

HOSE LIMITS (Table 1, summarized)

Maintain 60–100 psi while flowing ≥200 LPM per user. Approved total hose length: Hansen HK coupling (26369 series) — up to 150 ft / 1 segment. Foster, standard Hansen, Schrader, and CEJN coupling series — up to 300 ft / up to 3 segments. Always test couplings match your respirator equipment before relying on them in the field.

MAINTENANCE AFTER EACH USE

- Damp-sponge dirt from cart, cylinders, and fittings — don't obscure warning labels.
- Perform Regular Operational Inspection (alarm test, coupling check, leak check).
- Install two fully charged cylinders before returning to service.
- Store in a cool, dry area.
- Tag out and remove from service immediately if any damage, leak, or alarm failure is found.

This is a condensed field/training reference, not a substitute for the full Scott Safety O&M; manual (P/N 595269-01). Full manual: fedfiretrainingsd.org — search "Scott TRC-1 air cart manual." Scott Safety: 1-800-247-7257 · www.scottsafety.com

LABELED DIAGRAM



Diagram adapted from Scott Safety product illustration; labels translated to English for training reference.